

Practical 1 - Introduction to Survival Analysis.

Survival Analysis

London School of Hygiene & Tropical Medicine

Stata version (please see accompanying Stata do file)

SOLUTIONS

Datasets required: `pbcbase.dta`, `surv_data_practical1.dta`, `whitehall.dta`

Introduction

This practical is in three parts.

Part A. Looks at some fundamentals of working with survival data. Using `pbcbase`.

Part B. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `surv_data_practical1`.

Part C. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `whitehall`.

Aims

The aims of this session are:

- Understand how survival data are stored and set-up for analysis.
- Derive the formula for the maximum likelihood estimator of the hazard rate parameter λ from an exponential distribution.
- Be able to obtain the maximum likelihood estimate $\hat{\lambda}$ from data.

Part A: Using survival data

The first dataset we will use is called `pbcbase`. This is data from a multicentre, double-blind, clinical trial for the treatment of Primary Biliary Cirrhosis (PBC). There are 184 individuals in the data and 17 variables; the variables of interest in this session are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>datein</code>	Date person entered the study
<code>dateout</code>	Date of the end of follow-up due to either death or censoring
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years

Question 1

Open the `pbcbase` dataset and familiarise yourself with the key variables.

```
use "pbcbase.dta" , clear
```

Question 2

The `datein` and `dateout` variables are shown in Stata in the format (day,month,year). However, behind the scenes in Stata the dates are simply recorded as numbers. The number used by Stata to encode a particular date is as the number of days after 1st January 1960. So 1st January 1960 takes value 0, 2nd January 1960 takes value 1, etc. You can see the dates in this format by changing the format in which the dates are viewed:

```
list datein dateout in 1/5

format datein dateout %tg

list datein dateout in 1/5
```

To return to the more readable (day,month,year) format type

```
format datein dateout %td
```

You can find out more about how time is recorded and displayed in Stata using the help files for `datetime`, `format`, and `datetime_display_formats`.

Question 3

- How many people died in the dataset, and how many were censored?
- What was the earliest date of entry to the study? And the latest?
- What was the earliest date of exit from the study? And the latest?
- What happened to the first two people in the data (id=5 and id=23).

SOLUTION

```
tab d

summarize datein
display %td r(min)
display %td r(max)

summarize dateout
display %td r(min)
display %td r(max)

list in 1/2
```

- `d` is the event indicator, taking value 1 for individuals who die and 0 for individuals who are censored at their end of follow-up. Hence there were 96 deaths during the study follow-up, and 88 people were censored.
- The earliest date of entry was 3rd January 1980, and the latest was 25th December 1980.
- Exit dates ranged from 19th January 1980 to 26th June 1992.
- The person with id=5 joined the study on 19/2/1980 and died on 19/9/1981 (because `d=1`), which was a follow-up time of 1.58 years.
- The person with id=23 joined the study on 22/7/1980 and their censoring date (because `d=0`) was on 02/08/1989, which was a follow-up time of 9.03 years. That is, this person was known to be still alive after 9.03 years of follow-up.

Question 4

The fact that Stata stores dates as numbers makes calculating the time between two dates simple. To calculate the length of time each person was followed in the study (which we call `days _in_study`) we can type:

```
gen days_in_study = dateout - datein
```

Compare this new variable to the `time` variable to make sure this has done what you expect.

SOLUTION

- The `time` variable records the follow-up time (from `datein` to `dateout`) in years.
- When obtaining the difference between two dates we get that in days. This can be converted to years by dividing by 365.25 (this is a common way of handling leap years).

Question 5

- What do you think is the appropriate time origin in the PBC study?
- Before we can perform analyses of survival data in Stata it is necessary to use the `stset` command. This informs Stata what the time origin is and how the event of interest is indicated. One way to `stset` the data is using:

```
stset time, failure(d)
```

This method uses the follow-up time as the time scale (after the `stset`) and specifies the event indicator using the `failure()` option. As the `time` variable is in years, this will be the units in which time is used in the analysis.

Stata creates some new variables when you `stset` the data. Examine these new variables. What do they each represent? How are each of these new variables related to the variables you used in the `stset` command?

- Another way to `stset` the data is using:

```
stset dateout, origin(datein) failure(d)
```

Here the time of the end of follow-up is provided as the date each person left the study, and the options tell Stata the time origin (the date of joining the study, in this case), and their event indicator. Because the entry and exit dates (`datein` and `dateout`) are stored in number of days, the unit of analysis will be days.

Examine the `_t` variable, and its relationship with the `time` variable we created earlier.

- We can change the time unit using the `scale()` option. For example, to use years, rather than days we would type:

```
stset dateout, origin(datein) failure(d) scale(365.25)
```

SOLUTION

As this is a clinical trial, date from randomisation is the most natural timescale. Time is measured in days.

Part B: Fitting models to survival data

Question 1

Consider a study of n individuals. Let t_i denote the event or censoring time for individual i and let δ_i be the indicator of whether the person died (meaning t_i is their time of death) or was censored (meaning t_i is their censoring time, i.e. they were known to be alive at this time).

Write down the likelihood for survival data (including censoring) assumed to follow an exponential distribution with hazard λ . Derive a formula for the maximum likelihood estimate for λ .

SOLUTION

The likelihood is

$$L = \prod_{i=1}^n h(t_i)^{\delta_i} S(t_i) = \prod_{i=1}^n \lambda^{\delta_i} \exp(-\lambda t_i),$$

where $h(t)$ denotes the hazard at time t and $S(t)$ denotes the survival function at time t .

The log likelihood is therefore

$$l = \sum_{i=1}^n \delta_i \log \lambda - \lambda \sum_{i=1}^n t_i.$$

Differentiating the log likelihood with respect to λ gives

$$\frac{dl}{d\lambda} = \sum_{i=1}^n \delta_i / \lambda - \sum_{i=1}^n t_i.$$

The maximum likelihood estimator for λ is therefore $\hat{\lambda} = \sum_{i=1}^n \delta_i / \sum_{i=1}^n t_i$, which is the total number of events (deaths) divided by the total person-time of follow-up.

Question 2

We will now use the formula derived for $\hat{\lambda}$ in a simulated dataset called `surv_data_practical1.dta`. There are just two variables: `survtimes` contains the survival time for each participant, and `d` the outcome. In this dataset all participants experienced the event (`d=1`). The data were simulated for 100 individuals, with event times generated from an exponential distribution.

Load the data using

```
use "surv_data_practical1.dta" , clear
```

Using your formula from Part B, Question 1 calculate the maximum likelihood estimate for λ and interpret the estimate.

SOLUTION

```
summarize survtimes  
di r(N) / ( r(mean) * r(N) )
```

In these data, $\sum_{i=1}^n \delta_i = 100$ and $\sum_{i=1}^n t_i = 460.2566$. Thus the estimate of λ is $\hat{\lambda} = 0.217$. The interpretation is that the estimated hazard of mortality is 0.217 per year of follow-up. If we observed 1000 people for 1 year we would expect to see 217 deaths. If we observed 2000 people for 6 months we would expect to see 217 deaths. If we followed 500 individuals for 2 years we would expect to see 217 deaths. Because we used an exponential model, the hazard is assumed to be the same at all times, i.e. it does not change over time.

Question 3

We will fit an exponential model to these data to confirm the maximum likelihood estimate for λ that we calculated above. Compare the results with what you found above.

In Stata use **stset** to declare the data to be survival data. An exponential model can be fitted using the **streg** command:

```
streg, distribution(exponential)
```

See what happens if you use the **nohr** option in **streg**.

SOLUTION

- The estimate of λ is 0.217, which is what we found in our by-hand calculation above.
- See Table 3.1 in the lecture notes.

Part C: Whitehall study data

The Whitehall Study is a cohort study, which was started in 1967 and is a cohort of male civil servants employed in various government departments in London. It has been used to study risk factors for mortality in men. We will focus on deaths due to coronary heart disease (CHD). The dataset contains 1677 individuals and 14 variables, with the variables used in this practical summarised below:

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>timebth</code>	Date of birth
<code>timein</code>	Date person entered the study
<code>timeout</code>	Date of the end of follow-up due to either death or censoring
<code>chd</code>	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease

Question 1

- Load the data.
- What is the latest date of the end the follow-up? How many people had their end of follow-up on this date? What could the explanation for this be?

SOLUTION

- The latest date of the end the follow-up is 30th January 1987, when 1275 people are listed as being censored. This is most likely the date on which the study closed and follow-up was stopped, or the date this dataset was extracted.

Question 2

This is an example of data where event times could be left-truncated, meaning that not everyone is observed from the time origin of interest. In this type of study we could be interested in the date of joining the study as the time origin, or the date of birth as the time origin, depending on our question.

In Stata, compare the following four ways of using `stset` for this data:

```
** appropach (i)
stset timeout, failure(chd) origin(timein)

** appropach (ii)
stset timeout, failure(chd) origin(timebth)

** appropach (iii)
stset timeout, failure(chd) origin(timebth) enter(timein)

** appropach (iv)
stset timeout, failure(chd) origin(timebth) enter(timein) scale(365.25)
```

What is the time scale being used in each case? Examine the variables created by Stata in each case.

SOLUTION

- Approach (i) uses time-in-study as the timescale. Approaches (ii) and (iii) use age as the time scale, but (ii) does not account for the delayed entry/left truncation, which is incorrect. In (ii) individuals are being considered to be ‘at risk’ before they joined the cohort, which we should not allow. If there is delayed entry we should account for that in the analysis. Approach (iv) is the same as (iii) except time is measured in years instead of days.